Building Machine Learning Models with TensorFlow

This article offers an insight into machine learning models and how TensorFlow can be used to build them.

Machine learning is a young technology and is characterised by software that learns from past experiences to improve a system's performance. It is the most efficient way to develop an algorithm that translates its input to the related output.

For example, consider a series of numbers denoting a gray scale image as an input. It is difficult to write an algorithm to label each and every object of an image. This is where machine learning is so useful. It provides a toolset to read and write software without describing every detail of the algorithm. If the developer leaves some ambivalent values, machine learning will automatically figure out the best output by itself.

These ambivalent values are also called parameters, and their description is referred to as a model. The user's work is to write an algorithm that observes past examples to figure out how to tune parameters to achieve the model in an efficient way.

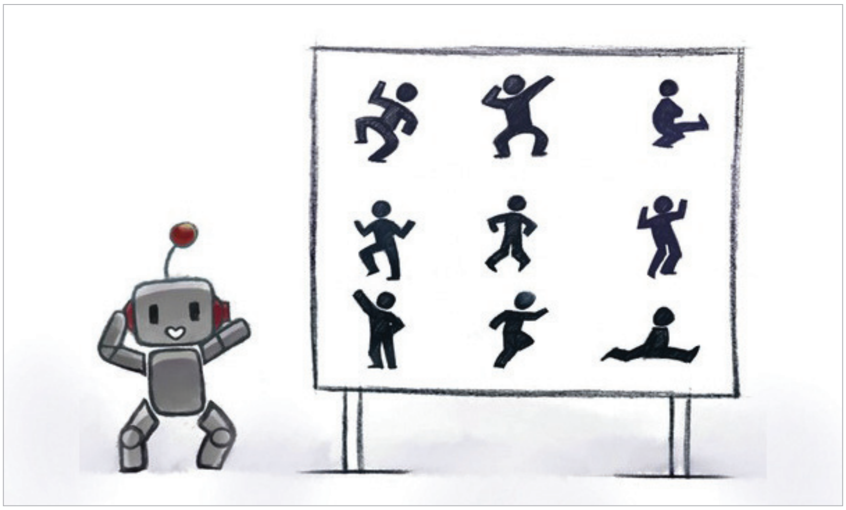

Figure 1: Machine learning fundamentals

Learning and inference

Machine learning algorithms are examined in two ways: learning and inference. The prime aim of the learning step is to describe the data, which is called a feature vector, and aggregate it in a model. The learning algorithm selects a model and actively searches for this model's parameters. This stage is more time consuming. The inference step moulds the model created in the learning step into an intelligent model.

Data representation, features and vector norms

The feature vector is a practical specification of data. To find out the accuracy of the real world data, the number of dimensions in the feature vector is included and the similarity is calculated by distance metrics (comparing similarity between the objects is an essential step in machine learning).

Let's take two feature vectors $a=(a_1, a_2,\ldots a_n)$ and $b=(b_1,b_2,\ldots b_n)$. The Euclidean distance || a-b || is calculated by:

$$\left(\sqrt{(a_1-b_1)^2+(a_2-b_2)^2+\cdots+(a_n-b_n)^2}\right.$$

The learning approach generally follows the structured models. Primarily, the data set needs to be transformed into a representation, which includes the everyday list of features and, finally, it can be used by the learning algorithm.

The inference process takes less time and can be faster when it works with real-time data. Once the process ends, the inference tests the model on new data.

Types of machine learning

There are three types of machine learning:

- Supervised learning
- Unsupervised learning
- Reinforcement learning

Supervised learning needs labelled data to implement a model. A model is just a function that labels data, including the past experiences. This is done with the help of past examples, which is denoted as a training data set.